EASY · ALGEBRA

Algebra

06

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

Q1 GRID-IN ALGEBRA

Maria plans to rent a boat. The boat rental costs \$60 per hour, and she will also have to pay for a water safety course that costs \$10. Maria wants to spend no more than \$280 for the rental and the course. If the boat rental is available only for a whole number of hours, what is the maximum number of hours for which Maria can rent the boat?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

In 2010, a swim club had a total of 35 swimmers, each classified as either advanced or intermediate. From 2010 to 2020, the number of advanced swimmers in the club increased by approximately 53%, and the number of intermediate swimmers in the club increased by approximately 44%. The total number of swimmers in the club increased by approximately 49%. Which equation best represents this situation, where a represents the number of advanced swimmers in the club in 2010 and b represents the number of intermediate swimmers in the club in 2010?

A. $1.53a + 1.49b = 35(1.44)$

B. $1.49a + 0.53b = 35(1.44)$

C. $1.53a + 1.44b = 35(1.49)$

D. $1.44a + 1.53b = 35(1.49)$

Q3 GRID-IN ALGEBRA

If $2x + 3 = 9$, what is the value of $6x$ - 1?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q4 ALGEBRA

The function h is defined by $hx = x + 200$. What is the value of $h50$?

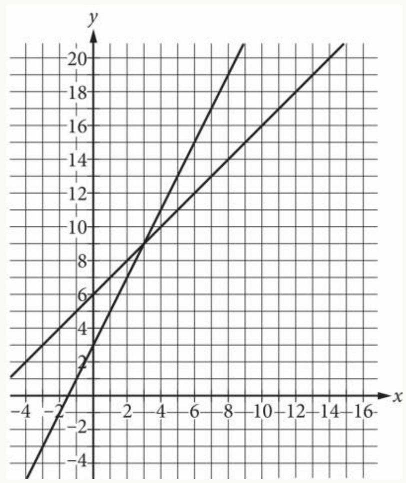
A. 200

B. 250

C. 10,000

D. 50,200

A system of two linear equations is graphed in the xy-plane below.



Which of the following points is the solution to the system of equations?

A. $(3,9)$

B. $(6,15)$

C. $(8,10)$

D. $(12,18)$

Q6

ALGEBRA

The total cost, in dollars, to rent a surfboard consists of a $\$ 25$ service fee and a $\$ 10$ per hour rental fee. A person rents a surfboard for t hours and intends to spend a maximum of $\$ 75$ to rent the surfboard. Which inequality represents this situation?

A. $10t \leq 75$

B. $10 + 25t \leq 75$

C. $25t \leq 75$

D. $25 + 10t \leq 75$

Q7

ALGEBRA

What is the y-intercept of the graph of $y = 34x + 81$ in the xy-plane?

A. $0, 81$

B. $0, 34$

C. $0, \text{ - } 34$

D. $0, \text{ - } 81$

Q8 ALGEBRA

One pound of grapes costs $2. At this rate, how many dollars will c pounds of grapes cost?

A. $2c$

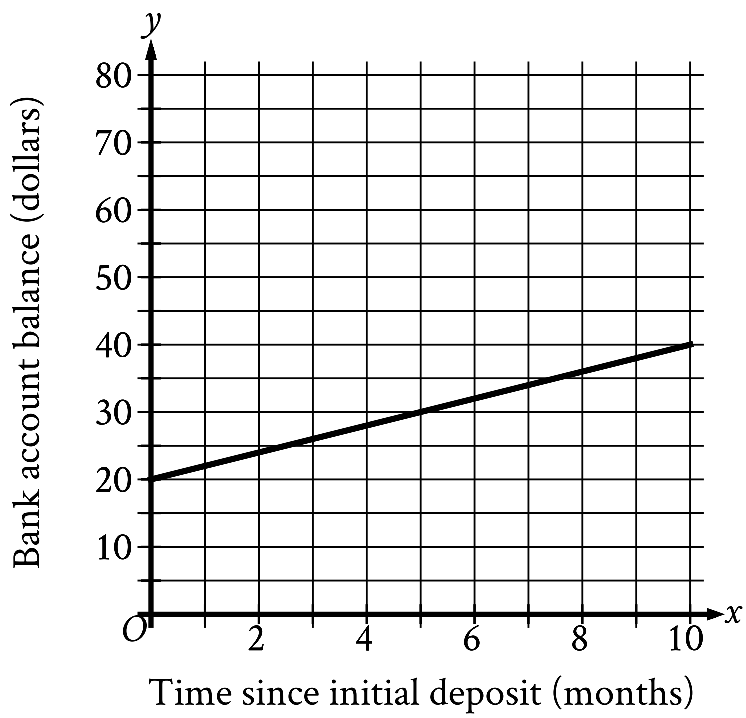
B. $2+c$

C. $\frac{2}{c}$

D. $\frac{c}{2}$

Q9

GRID-IN ALGEBRA



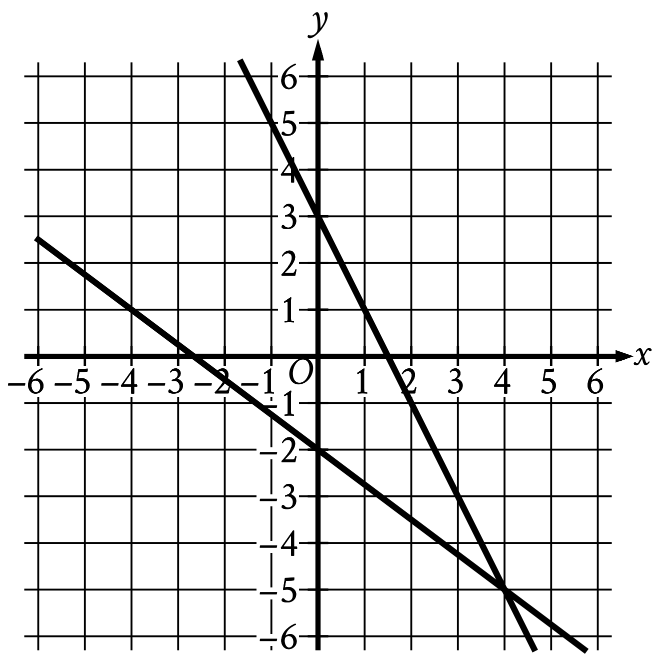
- The line slants gradually up from left to right.
- The line passes through the following points:
  - (0 comma 20)
  - (5 comma 30)
  - (10 comma 40)

A bank account was opened with an initial deposit. Over the next several months, regular deposits were made into this account, and there were no withdrawals made during this time. The graph of the function f shown, where $y = f(x)$, estimates the account balance, in dollars, in this bank account x months since the initial deposit. To the nearest whole dollar, what is the amount of the initial deposit estimated by the graph?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.



- For the first line in the system:
  - The line slants sharply down from left to right.
  - The line passes through the following points:
    - (0 comma 3)
    - (4 comma negative 5)
- For the second line in the system:
  - The line slants gradually down from left to right.
  - The line passes through the following points:
    - (0 comma negative 2)
    - (4 comma negative 5)

The graph of a system of linear equations is shown. What is the solution x, y to the system?

A. $4, -5$

B. $0, 3$

C. $0, -2$

D. $-2, 3$